Calculus(1)

WORKSHEET #1

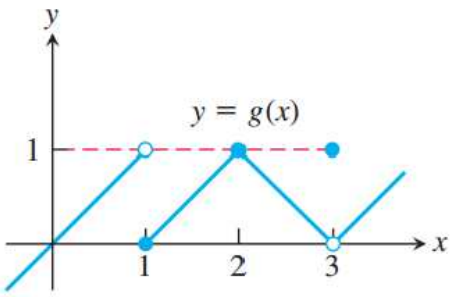
I D : _____

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※ Show the details of your work.

Ch1.

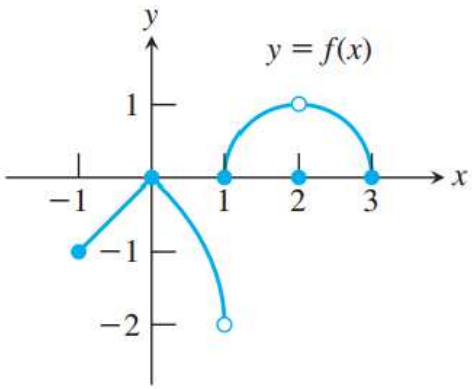
Q1. Find the following limits or explain why they do not exist.



(1) $\lim_{x\to 1} g(x)$

(2) $\lim_{x\to 2} g(x)$

(3) $\lim_{x\to 3} g(x)$



<figure for Q2>

Q2. Which of the following statements about the function $y = f(x)$ are true, and which are false?

(a) $\lim_{x\to 2} f(x)$ does not exist.

(b) $\lim_{x\to 2} f(x) = 2$

(c) $\lim_{x\to 1} f(x)$ does not exist.

(d) $\lim_{x\to c} f(x)$ exists at every $c \in (1,3)$.

(e) $\lim_{x\to c} f(x)$ exists for every $c \in (-1,1)$.

v	T	F
a		
b		
c		
d		
e		

Q3. Find the limit.

(1) $\lim_{x\to 0} \cos(x + \sin x)$

(2) $\lim_{x\to 3} \dfrac{x^2-9}{x^2+2x-3}$

(3) $\lim_{x\to -3} \dfrac{x^2-9}{x^2+2x-3}$

(4) $\lim_{x\to 1^+} \dfrac{x^2-9}{x^2+2x-3}$

(5) $\lim_{x\to 1}\left(\frac{1}{x-1}+\frac{1}{x^2-3x+2}\right)$

(6) $\lim_{x\to 4^+}\frac{4-x}{|4-x|}$

(7) $\lim_{h\to 0}\frac{\sqrt{5h+4}-2}{h}$

(8) $\lim_{x\to -1}\frac{\sqrt{x^2+8}-3}{x+1}$

(9) $\lim_{x\to 0}\frac{\frac{1}{x-1}+\frac{1}{x+1}}{x}$

(10) $\lim_{x\to 4}\frac{4-x}{5-\sqrt{x^2+9}}$

Q4. Use the Intermediate Value Theorem to show that there is a root of the equation.

(1) $x^3-3x-1=0$

(2) $2x^3-2x^2-2x+1=0$

(3) $\sqrt{x}+\sqrt{1+x}=6$

(4) $x(x-1)^2=1$

(5) $x^5-x^3+3x=5$

Q5. Evaluate the difference quotient for the given function. Simplify your answer.

(1) $f(x)=4+3x-x^2$; $\dfrac{f(3+h)-f(3)}{h}$

(2) $f(x)=\dfrac{1}{x}$; $\dfrac{f(x)-f(a)}{x-a}$

(3) $f(x)=\dfrac{x+3}{x+1}$; $\dfrac{f(x)-f(1)}{x-1}$

Q6. Find the domain of the function.

(1) $f(x)=\dfrac{2x^3-5}{x^2+x-6}$

(2) $f(x)=\sqrt{3-x}-\sqrt{2+x}$

(3) $f(x)=\dfrac{x+3}{4-\sqrt{x^2-9}}$

(4) $f(x)=\dfrac{1}{|x-2|}$

Q7. Find the range of $y = 2 + \dfrac{x^2}{x^2+4}$.

Q8. Determine whether f is even, odd, or neither.

(1) $f(x) = x|x|$

(2) $f(x) = 1 + 3x^2 - x^4$

(3) $f(x) = 1 + 3x^3 - x^5$

(4) $f(x) = \dfrac{1+3x^2-x^4}{1-3x^3-x^5}$

(5) $f(x) = \dfrac{1+3x^2-x^4}{3x^3-x^5}$

Q9. Find the functions (1) $f \circ g$, (2) $g \circ f$, (3) $f \circ f$, and (4) $g \circ g$ and their domains, where

(a) $f(x) = x + \dfrac{1}{x}$ and $g(x) = \dfrac{x+1}{x+2}$.

(b) $f(x) = \sqrt{x}$ and $g(x) = \sqrt{x^2 - 3x - 4}$.